Colors and Gases Could Help Scientists Find Tiny Alien Life

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If life exists on other worlds, it is probably tiny, one-celled life called **microbes**. Scientists are studying tough **microbes** on Earth to learn what clues might reveal life somewhere else.

One place they are looking is Venus. Its surface is hot enough to melt lead, so life as we know it could not survive there. But high in Venus's clouds, 48 to 60 kilometers (30 to 37 miles) up, temperatures and pressures are similar to Earth's. Scientists have spotted possible traces of gases such as oxygen, **methane** and phosphine there. In 2021, researchers at the Massachusetts Institute of Technology suggested that **microbes** might survive in those clouds by drying out when they sink into hotter layers, then reviving in cooler, wetter ones.

One of those researchers, Noelle Bryan, had collected **microbes** from Earth's **stratosphere**, a harsh layer of air high above the ground. Those **microbes** seemed to survive extreme drying.

Some of Bryan's bacteria make colorful **pigments** that protect them from harmful **ultraviolet** light. Lígia Coelho, an **astrobiologist** at Cornell University, found that these **pigments** absorb or reflect certain colors of light. Future telescopes might look for similar color patterns on planets around distant stars.

Mars is another target. Andrew Schuerger of the University of Florida and Rachel Harris of NASA and Harvard found a microbe called Methanosarcina barkeri that can live without oxygen and still make **methane** gas at low, Mars-like air pressure. Rovers have detected **methane** on Mars. Schuerger says a rover might only need to drill about 20 centimeters, or 8 inches, to reach Martian **microbes**, if they exist.

NASA scientist Lynn Rothschild even wonders whether Martian windstorms might carry **microbes** around, the way a fence catches blowing leaves.

Word Treasure

microbes -- Living things so small you need a microscope to see them.
stratosphere -- A high layer of Earth's atmosphere, above where airplanes usually fly.
pigments -- Natural substances that give something its color.
ultraviolet -- A kind of invisible light from the Sun that can harm living cells.
astrobiologist -- A scientist who studies whether life could exist beyond Earth.
methane -- A gas that some microbes make; it can be a sign of life.

Background you need

Astrobiology is the science of looking for life beyond Earth, and scientists think the first alien life we find might be tiny microbes.

Venus has a scorching surface, but its upper clouds are milder, which is why some scientists wonder if microbes could survive there.

Mars has thin air and no oxygen in its atmosphere, so scientists study tough Earth microbes to see what could survive there.

Telescopes and rovers can detect gases and colors from far away, which may give clues about whether life is present.

Break it down

LEAD: Earth microbes give clues for finding tiny alien life
KEY IDEA: Life elsewhere is probably one-celled microbes
+- VENUS: Mild cloud layers may host microbes that survive drying out
+- DISTANT PLANETS: Pigment colors could be spotted by future telescopes
EVIDENCE: Stratosphere microbes survive drying; pigments reflect certain colors of light
MARS: Methanosarcina barkeri makes methane at low pressure; rovers have found methane
OPEN QUESTION: Could Martian windstorms carry microbes, and could a rover catch them?

Why it matters

Studying tough microbes on Earth helps scientists decide what signs to look for on other worlds. It could shape how future telescopes and rovers search for life.

Test what you remember

1. Why can't life as we know it survive on Venus's surface?

- (A) It has too much water
- (B) It is covered in ice
- (C) It is hot enough to melt lead
- (D) It has no clouds

2. How high above Venus's surface are the clouds that interest scientists?

- (A) 500 kilometers
- (B) 1 to 2 kilometers
- (C) 48 to 60 kilometers
- (D) Only a few meters

3. Where did Noelle Bryan collect tough microbes?

- (A) Deep in the ocean
- (B) In Earth's stratosphere
- (C) On Mars
- (D) Inside a volcano

4. What do some bacteria's colorful pigments do?

- (A) Help them fly
- (B) Protect them from harmful ultraviolet light
- (C) Turn them into gas
- (D) Make them taste sweet

5. What can the microbe Methanosarcina barkeri do at low, Mars-like pressure?

- (A) Glow in the dark
- (B) Make methane gas
- (C) Freeze water
- (D) Make oxygen

6. How deep might a rover need to drill to reach Martian microbes, according to Schuerger?

- (A) It cannot be done
- (B) About 20 centimeters
- (C) About 2 kilometers
- (D) About 20 meters

Answer key (try the quiz first!): 1: C 2: C 3: B 4: B 5: B 6: B

Questions to think about

1. MIT researchers including Noelle Bryan: Suggested microbes might survive in Venus's clouds by drying out and reviving.

2. Lígia Coelho, Cornell astrobiologist: Hopes to look for pigment color patterns on distant planets.

3. Andrew Schuerger, University of Florida: Says a rover might only need to drill about 20 centimeters to reach Martian microbes, if they exist.

4. Lynn Rothschild, NASA: Wonders whether Martian windstorms might carry microbes from place to place.

MY THOUGHTS (at least 20 words):